Controlled turbulence-sensitivity study: error of the analytic averages versus σ_X^2 at fixed distance

Computed from the repository `Anand-Ambastha/tf-qkd-private`

22 October 2026

1 Design of the study

What is varied. For each fixed total distance $L \in \{80, 120, 160\}$ km the variance of the log-transmittance $X = \ln(\eta/\eta_0)$, $\eta = \sqrt{\eta_A \eta_B}$, is imposed on a grid $\sigma_X^2 \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6,$ independently of L. **This is a controlled sensitivity study.** The imposed σ_X^2 values are not claimed to correspond to atmospheric conditions realisable at that distance; the distance enters only through the deterministic transmittance $\eta_0 = \sqrt{\eta_{\mathrm{FSO}}(L)}$ (0.0393, 0.00757, 0.00175 at 80, 120, 160 km, i.e. the operating point on the key-rate curve). Where the channel model does tie σ_X^2 to the atmosphere this is stated below.

Three channel models.

- **LN**: $X \sim \mathcal{N}(-\sigma_X^2, \sigma_X^2)$ (manuscript model; $\sigma_X^2 = \sigma_R^2(L/2)/2$ per arm pair).
- **GG**: independent Gamma–Gamma arms, $X = \frac{1}{2}(\ln I_A + \ln I_B)$, $\mathrm{Var}(X) = c_2 = \frac{1}{2}[\psi'(\alpha) + \psi'(\beta)]$ with (α, β) from the plane-wave Andrews–Phillips formulas (exponent 6/5 on the Rytov variance, as in the repository code). For each target σ_X^2 the per-arm Rytov variance is solved from $c_2(\sigma_R^2) = \sigma_X^2$ on the rising branch. *c_2 is bounded:* its maximum over all σ_R^2 is 0.858 (at $\sigma_R^2 \approx 52$), because Gamma–Gamma saturates; this fixes the upper end of the grid. The grid values $\geq$ about 0.85 would correspond to saturated moderate/strong turbulence.
- **GG+BB**: as GG, with turbulence-induced long-term beam broadening (which contains beam wander) per arm, $\eta_0 \to \eta_0 f_{\mathrm{bb}}$, $W_{\mathrm{LT}}^2 = W^2(1 + 1.63\,\sigma_R^{12/5}\Lambda)$, using the σ_R^2 solved for GG. Broadening therefore changes the operating point, not the fading statistics. In this study σ_R^2 and L are independent, so f_{bb}^2 is that of an arm of length $L/2$ with the imposed σ_R^2 (listed in the tables); in a real atmosphere both follow from C_n^2.

Approximations compared (all with the repository's analytic derivatives, Appendices A–H of the manuscript): *conventional/mean* $\mathbb{E}[Q] \simeq Q(\bar{\eta}) + \frac{1}{2}Q''(\bar{\eta})\mathrm{Var}(\eta)$ with $\bar{\eta} = \mathbb{E}[\eta]$, $\mathrm{Var}(\eta) = \eta_0^2(1 - m_1^2)$, $m_1 = \mathbb{E}[e^X]$; *2nd-order median* $Q(\eta_m) + \frac{c_2}{2}\Theta^2 Q$; *4th-order median* $Q(\eta_m) + \frac{c_2}{2}\Theta^2 Q + \frac{c_3}{6}\Theta^3 Q + \frac{c_4 + 3c_2^2}{24}\Theta^4 Q$, all at $\eta_m = \eta_0 e^{c_1}$ (cumulants $c_n = 2^{1-n}\kappa_n$ for GG, $(-\sigma_X^2, \sigma_X^2, 0, 0)$ for LN).

References (independent of the formulas). LN: Simpson quadrature in X over $\mu \pm 12\sigma$ (40001 points). GG: numerical convolution of the two arms' Bessel-K log-densities (6001 points each; density norm $= 1 \pm 5 \times 10^{-10}$ in every case). The transmittance is capped at 1 in the references; the probability of $\eta > 1$ is at most 3.2×10^{-6} in all cases, so the cap does not affect the comparison. Errors are $100(\mathrm{approx} - \mathrm{ref})/\mathrm{ref}$.

Where this study touches realistic conditions. For the weak class $C_n^2 = 10^{-17}$ the actual values at $L = 80, 120, 160$ km are $\sigma_X^2 = 0.086$, 0.181, 0.307 (LN, manuscript Table) and $c_2 = 0.084$, 0.166, 0.258 (GG). For the moderate class the Gamma–Gamma channel is saturated, $c_2 = 0.84$–0.86 at all three distances (top of the grid), whereas the log-normal model would give $\sigma_X^2 = 8.6$–31, far outside the tested range. LN results above $\sigma_X^2 = 0.8$ were not computed.

2 Turbulence-sensitivity tables

Tables 3–5 give all $3 \times 3 \times 12$ cases. Table 1 condenses the key rate; Table 2 the maximum errors of e_X and e_Z.

Table 1: Secret key rate R: σ_X^2 at which |error| first exceeds 0.1% / 1% (log-linear interpolation on the grid; > 0.8 = not reached within the grid), and the maximum |error| [%] over $0.005 \le \sigma_X^2 \le 0.8$.

model	L [km]	mean: 0.1% / 1% / max	2nd: 0.1% / 1% / max	4th: 0.1% / 1% / max
LN	80	0.56 / > 0.8 / 0.24	0.08 / 0.25 / 8.12	0.28 / 0.65 / 1.71
LN	120	0.35 / > 0.8 / 0.70	0.09 / 0.28 / 7.16	0.34 / 0.78 / 1.06
LN	160	0.24 / 0.65 / 1.70	0.08 / 0.27 / 7.87	0.35 / 0.78 / 1.07
GG	80	> 0.8 / > 0.8 / 0.04	0.12 / 0.47 / 1.40	0.36 / 0.69 / 1.74
GG	120	0.63 / > 0.8 / 0.12	0.15 / > 0.8 / 0.89	0.36 / 0.70 / 1.63
GG	160	0.36 / > 0.8 / 0.27	0.15 / > 0.8 / 0.93	0.34 / 0.68 / 1.80
GG+BB	80	> 0.8 / > 0.8 / 0.09	0.13 / 0.51 / 1.11	0.36 / 0.70 / 1.63
GG+BB	120	0.56 / > 0.8 / 0.21	0.15 / > 0.8 / 0.87	0.36 / 0.70 / 1.72
GG+BB	160	0.35 / > 0.8 / 0.32	0.15 / 0.65 / 1.03	0.34 / 0.68 / 1.92

Table 2: Maximum |error| [%] of e_X and e_Z over $0.005 \le \sigma_X^2 \le 0.8$.

		e_X			e_Z		
model	L [km]	mean	2nd	4th	mean	2nd	4th
LN	80	0.000	0.025	0.003	0.485	0.069	0.011
LN	120	0.004	0.154	0.019	1.009	0.075	0.003
LN	160	0.060	0.580	0.059	1.857	0.209	0.009
GG	80	0.128	0.073	0.025	0.305	0.097	0.043
GG	120	0.637	0.360	0.111	0.718	0.260	0.044
GG	160	2.420	1.317	0.338	1.508	0.557	0.056
GG+BB	80	0.423	0.240	0.076	0.574	0.204	0.038
GG+BB	120	1.625	0.899	0.251	1.207	0.448	0.059
GG+BB	160	4.680	2.413	0.453	2.158	0.734	0.034

3 Error versus turbulence strength

Figures 1–3: panels (a) e_X, (b) e_Z, (c) R; colour/marker = approximation, line style = distance; absolute error on a logarithmic axis (signed values are in the tables).

4 Answers established by the data

1. Does the approximation error increase with turbulence strength? Yes, in general. Of the 81 (model, L, quantity, approximation) curves, 72 increase monotonically in |error| over the grid and 9 do not. The exceptions are the 2nd-order key-rate error of GG and GG+BB at all three distances (six curves; |error| peaks at $\sigma_X^2 \approx 0.6$–0.7 and then falls slightly, 1.40 → 1.26% at 80 km for GG),

Table 3: Turbulence-sensitivity table, Log-normal. Relative error [%] of the turbulence-averaged e_X, e_Z, R for the conventional mean-transmittance (delta) approximation, the 2nd-order and the 4th-order median formula, against the independent quadrature reference, at fixed L and imposed σ_X^2. σ_R^2: per-arm Rytov variance implied by $\sigma_X^2 = \sigma_R^2/2$ (manuscript convention), not an independent input.

				e_X			e_Z			R		
L [km]	σ_X^2	σ_R^2	f_{bb}^2	mean	2nd	4th	mean	2nd	4th	mean	2nd	4th
80	0.005	0.01	–	0	0	0	0	0	0	0	−0.0005	0
	0.01	0.02	–	0	0	0	0	0	0	0	−0.0018	0
	0.02	0.04	–	0	0	0	+0.0002	+0.0001	0	+0.0001	−0.0072	0
	0.05	0.1	–	0	0	0	+0.0011	+0.0007	0	+0.0006	−0.044	−0.0007
	0.1	0.2	–	0	0	0	+0.0047	+0.0026	0	+0.0025	−0.173	−0.0055
	0.2	0.4	–	0	−0.0003	0	+0.020	+0.0095	+0.0003	+0.010	−0.661	−0.041
	0.3	0.6	–	0	−0.0011	−0.0001	+0.048	+0.019	+0.0010	+0.025	−1.42	−0.128
	0.4	0.8	–	0	−0.0028	−0.0002	+0.092	+0.030	+0.0022	+0.046	−2.41	−0.282
	0.5	1	–	0	−0.0055	−0.0004	+0.154	+0.042	+0.0039	+0.077	−3.61	−0.513
	0.6	1.2	–	0	−0.0098	−0.0010	+0.238	+0.052	+0.0060	+0.117	−4.97	−0.827
	0.7	1.4	–	+0.0001	−0.016	−0.0018	+0.346	+0.062	+0.0085	+0.171	−6.49	−1.23
	0.8	1.6	–	+0.0001	−0.025	−0.0032	+0.485	+0.069	+0.011	+0.240	−8.12	−1.71
120	0.005	0.01	–	0	0	0	0	0	0	0	−0.0004	0
	0.01	0.02	–	0	0	0	+0.0001	0	0	+0.0001	−0.0015	0
	0.02	0.04	–	0	0	0	+0.0004	0	0	+0.0003	−0.0059	0
	0.05	0.1	–	0	−0.0003	0	+0.0024	−0.0001	0	+0.0018	−0.036	−0.0004
	0.1	0.2	–	0	−0.0011	0	+0.0099	−0.0004	0	+0.0073	−0.142	−0.0029
	0.2	0.4	–	0	−0.0048	−0.0002	+0.042	−0.0020	0	+0.031	−0.549	−0.022
	0.3	0.6	–	+0.0001	−0.012	−0.0006	+0.102	−0.0054	0	+0.073	−1.19	−0.071
	0.4	0.8	–	+0.0003	−0.024	−0.0015	+0.194	−0.011	−0.0001	+0.138	−2.05	−0.159
	0.5	1	–	+0.0006	−0.043	−0.0034	+0.323	−0.021	−0.0004	+0.228	−3.09	−0.297
	0.6	1.2	–	+0.0012	−0.069	−0.0065	+0.497	−0.034	−0.0009	+0.349	−4.30	−0.490
	0.7	1.4	–	+0.0021	−0.105	−0.011	+0.723	−0.052	−0.0018	+0.504	−5.66	−0.743
	0.8	1.6	–	+0.0036	−0.154	−0.019	+1.01	−0.075	−0.0033	+0.700	−7.16	−1.06
160	0.005	0.01	–	0	0	0	0	0	0	0	−0.0004	0
	0.01	0.02	–	0	0	0	+0.0002	0	0	+0.0002	−0.0016	0
	0.02	0.04	–	0	−0.0002	0	+0.0007	−0.0001	0	+0.0007	−0.0062	0
	0.05	0.1	–	0	−0.0011	0	+0.0046	−0.0005	0	+0.0042	−0.039	−0.0003
	0.1	0.2	–	+0.0002	−0.0045	−0.0001	+0.019	−0.0021	0	+0.017	−0.152	−0.0027
	0.2	0.4	–	+0.0009	−0.020	−0.0006	+0.081	−0.0090	−0.0001	+0.073	−0.589	−0.021
	0.3	0.6	–	+0.0025	−0.050	−0.0022	+0.193	−0.022	−0.0005	+0.176	−1.28	−0.067
	0.4	0.8	–	+0.0057	−0.099	−0.0057	+0.364	−0.041	−0.0011	+0.332	−2.21	−0.153
	0.5	1	–	+0.011	−0.170	−0.012	+0.604	−0.069	−0.0022	+0.551	−3.36	−0.288
	0.6	1.2	–	+0.021	−0.271	−0.022	+0.925	−0.105	−0.0039	+0.843	−4.69	−0.481
	0.7	1.4	–	+0.036	−0.405	−0.038	+1.34	−0.152	−0.0061	+1.22	−6.21	−0.739
	0.8	1.6	–	+0.060	−0.580	−0.059	+1.86	−0.209	−0.0089	+1.70	−7.87	−1.07

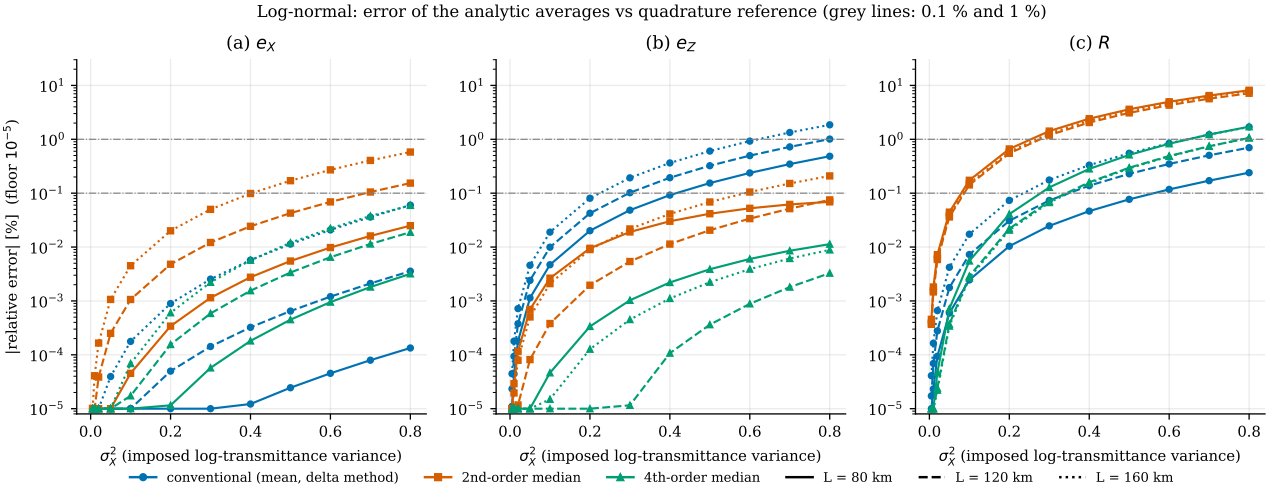


Figure 1: Log-normal channel.

Table 4: Turbulence-sensitivity table, Gamma–Gamma (per arm). Relative error [%] of the turbulence-averaged e_X, e_Z, R for the conventional mean-transmittance (delta) approximation, the 2nd-order and the 4th-order median formula, against the independent quadrature reference, at fixed L and imposed σ_X^2. σ_R^2: per-arm Rytov variance solved so that $\mathrm{Var}(X) = \sigma_X^2$; f_{bb}^2: end-to-end broadening factor for an arm of length $L/2$ with that σ_R^2 (shown for reference only, not applied in this table).

				e_X			e_Z			R		
L [km]	σ_X^2	σ_R^2	f_{bb}^2	mean	2nd	4th	mean	2nd	4th	mean	2nd	4th
80	0.005	0.00999	1.000	0	0	0	0	0	0	0	−0.0002	0
	0.01	0.02	1.000	0	0	0	0	0	0	0	−0.0008	0
	0.02	0.04	1.000	0	0	0	−0.0001	0	0	0	−0.0031	0
	0.05	0.101	1.000	−0.0002	−0.0001	0	−0.0004	+0.0001	0	+0.0003	−0.019	+0.0002
	0.1	0.207	1.000	−0.0008	−0.0003	0	−0.0018	+0.0001	−0.0001	+0.0011	−0.073	+0.0020
	0.2	0.449	1.000	−0.0038	−0.0015	−0.0001	−0.0081	−0.0001	−0.0005	+0.0045	−0.265	+0.016
	0.3	0.752	1.000	−0.0094	−0.0040	−0.0005	−0.021	−0.0020	−0.0016	+0.0097	−0.532	+0.056
	0.4	1.15	1.000	−0.019	−0.0085	−0.0014	−0.043	−0.0070	−0.0038	+0.016	−0.827	+0.143
	0.5	1.72	1.000	−0.033	−0.016	−0.0032	−0.077	−0.017	−0.0079	+0.023	−1.10	+0.308
	0.6	2.59	1.000	−0.054	−0.028	−0.0067	−0.129	−0.033	−0.015	+0.030	−1.31	+0.588
	0.7	4.19	1.000	−0.085	−0.045	−0.013	−0.201	−0.059	−0.026	+0.036	−1.40	+1.03
	0.8	8.77	1.000	−0.128	−0.073	−0.025	−0.305	−0.097	−0.043	+0.038	−1.26	+1.74
120	0.005	0.00999	1.000	0	0	0	0	0	0	0	−0.0001	0
	0.01	0.02	1.000	0	0	0	0	0	0	0	−0.0005	0
	0.02	0.04	1.000	−0.0002	−0.0001	0	−0.0002	−0.0001	0	+0.0002	−0.0021	0
	0.05	0.101	1.000	−0.0010	−0.0004	0	−0.0011	−0.0004	0	+0.0009	−0.013	+0.0003
	0.1	0.207	1.000	−0.0043	−0.0019	−0.0001	−0.0045	−0.0018	0	+0.0037	−0.050	+0.0022
	0.2	0.449	1.000	−0.019	−0.0086	−0.0005	−0.020	−0.0081	−0.0003	+0.014	−0.185	+0.017
	0.3	0.752	1.000	−0.048	−0.022	−0.0021	−0.051	−0.020	−0.0012	+0.031	−0.375	+0.059
	0.4	1.15	1.000	−0.096	−0.046	−0.0060	−0.104	−0.040	−0.0031	+0.051	−0.583	+0.147
	0.5	1.72	1.000	−0.169	−0.084	−0.014	−0.186	−0.070	−0.0069	+0.074	−0.767	+0.308
	0.6	2.59	1.000	−0.276	−0.143	−0.030	−0.307	−0.114	−0.014	+0.095	−0.885	+0.574
	0.7	4.19	1.000	−0.426	−0.230	−0.059	−0.478	−0.175	−0.025	+0.113	−0.892	+0.990
	0.8	8.77	1.000	−0.637	−0.360	−0.111	−0.718	−0.260	−0.044	+0.120	−0.684	+1.63
160	0.005	0.00999	1.000	0	0	0	0	0	0	0	−0.0001	0
	0.01	0.02	1.000	−0.0002	−0.0001	0	−0.0001	0	0	+0.0001	−0.0005	0
	0.02	0.04	1.000	−0.0007	−0.0003	0	−0.0004	−0.0002	0	+0.0004	−0.0022	0
	0.05	0.101	1.000	−0.0043	−0.0018	0	−0.0025	−0.0011	0	+0.0022	−0.013	+0.0003
	0.1	0.207	1.000	−0.018	−0.0077	−0.0002	−0.010	−0.0046	−0.0001	+0.0088	−0.051	+0.0025
	0.2	0.449	1.000	−0.079	−0.035	−0.0021	−0.046	−0.020	−0.0006	+0.034	−0.191	+0.020
	0.3	0.752	1.000	−0.197	−0.091	−0.0081	−0.116	−0.049	−0.0023	+0.073	−0.390	+0.067
	0.4	1.15	1.000	−0.390	−0.185	−0.022	−0.234	−0.095	−0.0061	+0.121	−0.608	+0.165
	0.5	1.72	1.000	−0.680	−0.333	−0.052	−0.413	−0.164	−0.013	+0.173	−0.802	+0.343
	0.6	2.59	1.000	−1.09	−0.554	−0.105	−0.672	−0.261	−0.025	+0.221	−0.926	+0.637
	0.7	4.19	1.000	−1.66	−0.872	−0.195	−1.03	−0.390	−0.040	+0.257	−0.931	+1.09
	0.8	8.77	1.000	−2.42	−1.32	−0.338	−1.51	−0.557	−0.056	+0.266	−0.702	+1.80

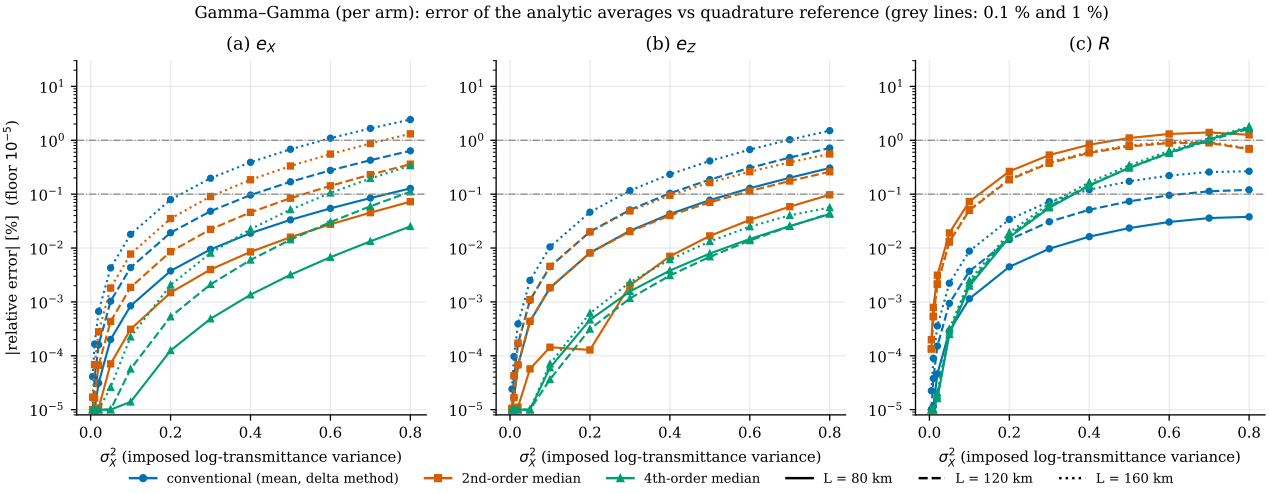


Figure 2: Gamma–Gamma channel (per arm).

Table 5: Turbulence-sensitivity table, Gamma–Gamma + beam wander (per arm). Relative error [%] of the turbulence-averaged e_X, e_Z, R for the conventional mean-transmittance (delta) approximation, the 2nd-order and the 4th-order median formula, against the independent quadrature reference, at fixed L and imposed σ_X^2. σ_R^2: per-arm Rytov variance solved so that $\mathrm{Var}(X) = \sigma_X^2$; f_{bb}^2: end-to-end broadening factor for an arm of length $L/2$ with that σ_R^2 (applied in this model; $= 1$ and unused in the plain GG model).

				e_X			e_Z			R		
L [km]	σ_X^2	σ_R^2	f_{bb}^2	mean	2nd	4th	mean	2nd	4th	mean	2nd	4th
80	0.005	0.00999	0.999	0	0	0	0	0	0	0	−0.0002	0
	0.01	0.02	0.997	0	0	0	0	0	0	0	−0.0008	0
	0.02	0.04	0.993	0	0	0	−0.0001	0	0	0	−0.0031	0
	0.05	0.101	0.978	−0.0002	−0.0001	0	−0.0004	+0.0001	0	+0.0003	−0.019	+0.0002
	0.1	0.207	0.949	−0.0009	−0.0003	0	−0.0018	+0.0001	−0.0001	+0.0012	−0.072	+0.0020
	0.2	0.449	0.878	−0.0040	−0.0016	−0.0001	−0.0084	−0.0005	−0.0004	+0.0049	−0.258	+0.016
	0.3	0.752	0.790	−0.011	−0.0046	−0.0005	−0.022	−0.0033	−0.0014	+0.011	−0.508	+0.056
	0.4	1.15	0.685	−0.023	−0.010	−0.0016	−0.047	−0.011	−0.0034	+0.020	−0.771	+0.143
	0.5	1.72	0.560	−0.044	−0.022	−0.0041	−0.090	−0.025	−0.0068	+0.031	−0.990	+0.305
	0.6	2.59	0.415	−0.084	−0.043	−0.0098	−0.161	−0.051	−0.012	+0.046	−1.11	+0.575
	0.7	4.19	0.254	−0.167	−0.090	−0.024	−0.287	−0.097	−0.021	+0.064	−1.07	+0.992
	0.8	8.77	0.087	−0.423	−0.240	−0.076	−0.574	−0.204	−0.038	+0.094	−0.746	+1.63
120	0.005	0.00999	0.999	0	0	0	0	0	0	0	−0.0001	0
	0.01	0.02	0.998	0	0	0	0	0	0	0	−0.0005	0
	0.02	0.04	0.995	−0.0002	−0.0001	0	−0.0002	−0.0001	0	+0.0002	−0.0021	0
	0.05	0.101	0.984	−0.0010	−0.0004	0	−0.0011	−0.0004	0	+0.0009	−0.013	+0.0003
	0.1	0.207	0.962	−0.0044	−0.0019	−0.0001	−0.0046	−0.0019	0	+0.0037	−0.050	+0.0022
	0.2	0.449	0.909	−0.020	−0.0090	−0.0006	−0.021	−0.0084	−0.0003	+0.015	−0.184	+0.017
	0.3	0.752	0.840	−0.053	−0.024	−0.0023	−0.054	−0.021	−0.0012	+0.032	−0.373	+0.060
	0.4	1.15	0.754	−0.110	−0.053	−0.0069	−0.112	−0.044	−0.0033	+0.056	−0.578	+0.148
	0.5	1.72	0.645	−0.210	−0.104	−0.017	−0.210	−0.080	−0.0075	+0.083	−0.758	+0.311
	0.6	2.59	0.509	−0.383	−0.197	−0.041	−0.369	−0.139	−0.016	+0.115	−0.871	+0.582
	0.7	4.19	0.340	−0.714	−0.383	−0.096	−0.637	−0.239	−0.032	+0.153	−0.870	+1.01
	0.8	8.77	0.133	−1.62	−0.899	−0.251	−1.21	−0.448	−0.059	+0.210	−0.659	+1.72
160	0.005	0.00999	0.999	0	0	0	0	0	0	0	−0.0001	0
	0.01	0.02	0.998	−0.0002	−0.0001	0	−0.0001	0	0	+0.0001	−0.0005	0
	0.02	0.04	0.996	−0.0007	−0.0003	0	−0.0004	−0.0002	0	+0.0004	−0.0022	0
	0.05	0.101	0.987	−0.0043	−0.0018	0	−0.0025	−0.0011	0	+0.0022	−0.013	+0.0003
	0.1	0.207	0.971	−0.018	−0.0078	−0.0002	−0.011	−0.0046	−0.0001	+0.0088	−0.051	+0.0025
	0.2	0.449	0.928	−0.082	−0.036	−0.0021	−0.047	−0.020	−0.0006	+0.035	−0.192	+0.020
	0.3	0.752	0.873	−0.210	−0.096	−0.0085	−0.121	−0.051	−0.0024	+0.076	−0.393	+0.068
	0.4	1.15	0.801	−0.432	−0.204	−0.024	−0.248	−0.101	−0.0064	+0.130	−0.617	+0.168
	0.5	1.72	0.707	−0.795	−0.388	−0.059	−0.453	−0.180	−0.014	+0.192	−0.822	+0.350
	0.6	2.59	0.582	−1.39	−0.697	−0.127	−0.770	−0.299	−0.026	+0.257	−0.970	+0.656
	0.7	4.19	0.414	−2.40	−1.24	−0.251	−1.27	−0.475	−0.034	+0.317	−1.03	+1.14
	0.8	8.77	0.182	−4.68	−2.41	−0.453	−2.16	−0.734	+0.030	+0.294	−1.03	+1.92

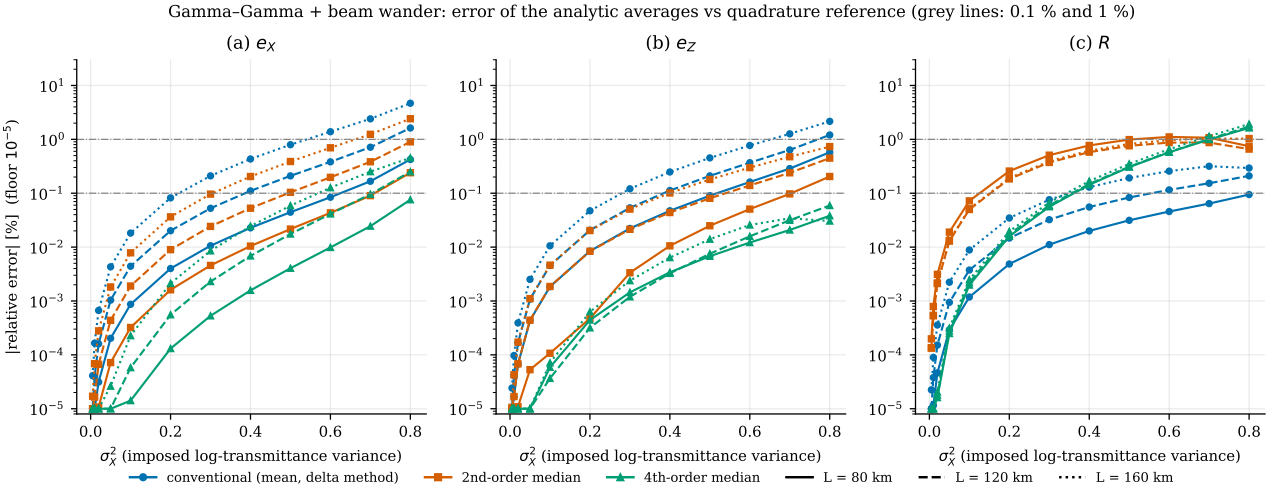


Figure 3: Gamma–Gamma channel with beam wander (long-term beam broadening).

the 2nd-order e_Z of GG at 80 km, and two small non-monotonicities at GG+BB, 160 km (mean-approximation R, 4th-order e_Z). The size and the ranking of the errors depend on the quantity and on the distance (Table 1, 2): errors grow with distance at fixed σ_X^2 for e_X and e_Z in the GG models (e.g. mean-approximation e_X error at $\sigma_X^2 = 0.8$: 0.13%, 0.64%, 2.4% at $80, 120, 160$ km), because the operating point moves to a smaller η_0 where the error rates vary faster in $\ln \eta$. For R the 4th-order error at $\sigma_X^2 = 0.8$ is nearly distance independent (1.6–1.9% in GG, 1.1–1.7% in LN).

2. When does the 2nd-order approximation become insufficient for the key rate? Taking 0.1% as the threshold for a "precise" value and 1% for a "usable" one: the 2nd-order key-rate error exceeds 0.1% at $\sigma_X^2 \approx 0.08$–0.09 for the log-normal channel and at 0.125–0.153 for Gamma–Gamma, with or without beam wander; it exceeds 1% at $\sigma_X^2 \approx 0.25$–0.28 (LN, all distances) and 0.47 (80 km, GG), 0.51 (GG+BB, 80 km), and 0.65 (GG+BB, 160 km); for GG at 120 and 160 km and GG+BB at 120 km it stays below 1% up to 0.8 (maxima 0.89, 0.93, 0.87%). The log-normal 2nd-order error reaches 7–8% at $\sigma_X^2 = 0.8$. Conclusion: for the key rate the 2nd-order formula stops being accurate (to 0.1%) at $\sigma_X^2 \approx 0.1$, and (to 1%) at ≈ 0.25 in log-normal and around 0.5 or not within the grid in Gamma–Gamma. At the realistic weak-class points ($\sigma_X^2 = 0.086$–0.307 LN, $c_2 = 0.084$–0.258 GG) the 2nd-order key-rate error, interpolated on the grid, is about 0.06, 0.14, 0.3% (GG, $80, 120, 160$ km) and 0.13, 0.5, 1.3% (LN).

3. Does the 4th-order approximation stay accurate over the tested range? It depends on the quantity.

- *e_X and e_Z:* yes. The 4th-order error stays below 0.46% everywhere and below 0.1% for all LN cases and for GG at 80 km; its maximum is 0.45% (e_X, GG+BB, 160 km) and 0.06% for e_Z. At $\sigma_X^2 = 0.8$ it is the best of the three approximations for e_Z in every case and for e_X in all GG and GG+BB cases and in LN at 160 km (in LN at 80 and 120 km the mean approximation is marginally better for e_X, $\leq 0.004\%$).

- *R:* the error is bounded ($\leq 1.9\%$ for $\sigma_X^2 \leq 0.8$) but it is *not* uniformly small. It exceeds 0.1% at $\sigma_X^2 \approx 0.28$–0.36 (LN 0.28–0.35, GG and GG+BB 0.34–0.36) and 1% at ≈ 0.65–0.78 (LN 0.65 at 80 km and 0.78 at 120 and 160 km; GG and GG+BB 0.68–0.70). In Gamma–Gamma the 4th-order error *overtakes* the 2nd-order one at $\sigma_X^2 \approx 0.7$ ($L = 120, 160$ km) and 0.8 (80 km, GG and GG+BB): adding the Θ^3, Θ^4 terms no longer improves the result, i.e. the cumulant series is asymptotic and the third and fourth cumulants are no longer small corrections there. In the log-normal channel the 4th order is better than the 2nd throughout (1.1–1.7% against 7–8% at 0.8).

- *Surprise in the data:* the conventional mean-transmittance approximation is the *most* accurate for R at $\sigma_X^2 = 0.8$ in all GG and GG+BB cases ($\leq 0.30\%$) and in LN at 80–120 km, although it is the worst for e_X and e_Z. This is not evidence of a controlled expansion: at these operating points $R(\eta)$ is locally close to a power law ($d \ln R / d \ln \eta = 1.05$–$1.13$ at 80–160 km in the repository), for which the curvature correction in η is small, and the delta-method error at 160 km in the log-normal model already grows to 1.7% at $\sigma_X^2 = 0.8$.

Beam wander. In this controlled setting beam wander only shifts η_0 by f_{bb} (from 0.999 to 0.09–0.18 at $\sigma_X^2 = 0.8$); the key rate itself drops by up to a factor 3.6 (2.8–3.6 at $\sigma_X^2 = 0.8$). The relative errors of R differ from the plain GG case by at most 0.13 percentage points. The errors of e_X and e_Z grow with the shift towards smaller η_0, mostly at 160 km (mean approximation e_X at $\sigma_X^2 = 0.8$: $2.4 \to 4.7\%$; 4th order: $0.34 \to 0.45\%$). The analytic formulas therefore keep their accuracy for the key rate with the beam-wander loss, and lose some for e_X with the conventional approximation, *provided* the shifted operating point stays on the smooth branch of the key-rate curve (all η_m here do; $R(\eta_m) \geq 6 \times 10^{-6}$).

5 Limits

Only $\sigma_X^2 \leq 0.8$ was tested (the Gamma–Gamma maximum is 0.858); log-normal behaviour above that, where the repository's validity limit $\sigma_R^2 = 1$ ($\sigma_X^2 = 0.5$) is already exceeded for $\sigma_X^2 > 0.5$, is not covered. The imposed values at fixed distance are a sensitivity device. Error values below $10^{-5}\%$ are plotted at a floor. Reproduction: `paper_analysis/scripts/sensitivity_study.py`, `sensitivity_thresholds.py`, `sensitivity_figures.py`, `sensitivity_make_tex.py` (they import the repository unchanged).